The Mathematical Association of Victoria

Trial Examination 2025

SPECIALIST MATHEMATICS

Trial Written Examination 1 - SOLUTIONS

Question 1 (3 marks)

$p(z) = 3z^4 - z^3 + z^2 - z - 2$.

One root of $p(z)$ is $-i$

$\therefore p(z) = (z-i)(z+i)(\text{quadratic})$

Divide to find the quadratic factor using $(z-i)(z+i) = z^2 + 1$ **1M**

$$\frac{3z^4 - z^3 + z^2 - z - 2}{z^2 + 1}$$

long division

$$\begin{array}{r|l}
 & \mathbf{3z^2 - z - 2} \\
z^2 + 0z + 1 & 3z^4 - z^3 + z^2 - z - 2 \\
 & 3z^4 + 0z^3 + 3z^2 \\
 & \quad -z^3 - 2z^2 - z \\
 & \quad -z^3 + 0z^2 - z \\
 & \qquad -2z^2 + 0z - 2 \\
 & \qquad -2z^2 + 0z - 2 \\
 & \qquad\qquad 0
\end{array}$$

OR

synthetic division

$$\begin{array}{r|rrrrr}
 & 3 & -1 & 1 & -1 & -2 \\
i & & 3i & -i+3 & 1+2i & -2 \\
\hline
 & 3 & 3i-1 & -i+2 & 2i & 0 \\
-i & & -3i & i & -2i & \\
\hline
 & \mathbf{3} & \mathbf{-1} & \mathbf{2} & \mathbf{0} &
\end{array}$$

Using either division method

$\therefore p(z) = (z-i)(z+i)(3z^2 - z - 2)$ **1M**

Giving

$p(z) = (z-i)(z+i)(3z+2)(z-1)$

Solutions to $p(z) = 0$ are $z = \pm i, z = -\frac{2}{3}, z = 1$ **1A**

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Question 2 (3 marks)

$$\frac{x}{\sqrt{9-y^2}}=\frac{1}{y}\frac{dx}{dy}$$

$$\frac{xy}{\sqrt{9-y^2}}=\frac{dx}{dy}$$

Separable variables

$$\int\left(\frac{1}{x}\right)dx=\int\left(\frac{y}{\sqrt{9-y^2}}\right)dy$$

Let $u=9-y^2 \therefore \frac{du}{dy}=-2y$

$$\log_e\left(|x|\right)=\int-\frac{1}{2}\left(\frac{-2y}{\sqrt{u}}\right)dy$$

$$\log_e\left(|x|\right)=\int-\frac{1}{2}\left(\frac{\frac{du}{dy}}{\sqrt{u}}\right)dy$$

$$\log_e\left(|x|\right)=-\frac{1}{2}\int\left(u^{-\frac{1}{2}}\right)du$$

$$\log_e\left(|x|\right)=-\frac{1}{2}\left(\frac{u^{\frac{1}{2}}}{\frac{1}{2}}\right)+c$$

$$\log_e\left(|x|\right)=-(9-y^2)^{\frac{1}{2}}+c$$

$\log_e\left(|x|\right)=-\sqrt{9-y^2}+c$ where c is a positive real constant `**1M**

Rearrange to find y in terms of x

$$\log_e\left(|x|\right)-c=-\sqrt{9-y^2}$$

For $x>0$. $\log_e\left(x\right)-c=-\sqrt{9-y^2}$

For $x=1$ when $y=2\sqrt{2}$

$$\log_e\left(1\right)-c=-\sqrt{9-\left(2\sqrt{2}\right)^2}$$

$$-c=-\sqrt{1}$$

$$\therefore c=1$$

Giving $\log_e\left(x\right)-1=-\sqrt{9-y^2}$

Square both sides

$$\left(\log_e\left(x\right)-1\right)^2=9-y^2$$

$y^2=9-\left(\log_e\left(x\right)-1\right)^2$ **1M**

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Gives $y=\pm\sqrt{9-\left(\log_e(x)-1\right)^2}$

Take positive solution

$y=\sqrt{9-\left(\log_e(x)-1\right)^2}$ **1A**

Question 3 (4 marks)

In the equation $2x^2y+3y^2-4x^2=1$, given $y=1$ gives $x=\pm1$

Find points $2x^2y+3y^2-4x^2=1$ where $y=1$

Gives $2x^2+3-4x^2=1\Rightarrow 2x^2=2$

gives $x=\pm1$

$\left(y\times4x+2x^2\times1\dfrac{dy}{dx}\right)+6y\dfrac{dy}{dx}-8x=0$ **1M**

$\dfrac{dy}{dx}\left(2x^2+6y\right)=8x-4xy$

$\therefore\dfrac{dy}{dx}=\dfrac{8x-4xy}{2x^2+6y}=\dfrac{4x-2xy}{x^2+3y}$

Given $y=1$ when $x=1$. $\therefore\dfrac{dy}{dx}=\dfrac{1}{2}$

Given $y=1$ when $x=-1$. $\therefore\dfrac{dy}{dx}=-\dfrac{1}{2}$ **1A**

Now $\dfrac{d^2y}{dx^2}=\dfrac{\left(x^2+3y\right)\left(4-(y\times2+2x\times\dfrac{dy}{dx})\right)-\left(4x-2xy\right)\left(2x+3\dfrac{dy}{dx}\right)}{\left(x^2+3y\right)^2}$

At $\dfrac{dy}{dx}=\dfrac{1}{2}$ and the point $(1,1)$

$\dfrac{d^2y}{dx^2}=\dfrac{(4)(4-3)-(2)\left(2+\dfrac{3}{2}\right)}{(4)^2}$

$\dfrac{d^2y}{dx^2}=-\dfrac{3}{16}$ **1A**

OR

At $\dfrac{dy}{dx}=-\dfrac{1}{2}$ and the point $(-1,1)$

$\dfrac{d^2y}{dx^2}=\dfrac{(4)(4-3)-(-4+2)\left(-2-\dfrac{3}{2}\right)}{(4)^2}$

$\dfrac{d^2y}{dx^2}=-\dfrac{3}{16}$ **1A**

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Question 4 (10 marks)

a. State the equations of asymptotes in the graph of $f(x)=\dfrac{2x^3}{x^2-4x+4}$

$$f(x)=\frac{2x^3}{x^2-4x+4}=2x+8+\frac{24x-32}{(x-2)^2}$$

$$\begin{array}{r|l} & \mathbf{2x+8} \\ \hline x^2-4x+4 & 2x^3+0x^2+0x+0 \\ & 2x^3-8x^2+8x \\ \hline & 8x^2-8x+0 \\ & 8x^2-32x+32 \\ \hline & \mathbf{24x-32} \end{array}$$

Oblique asymptote $y=2x+8$

Vertical asymptote $x=2$ **1A**

b. Axial intercepts of oblique asymptote.

$(-4,0)$ and $(0,8)$ **1A**

c. Stationary points

$$f(x)=\frac{2x^3}{x^2-4x+4}=\frac{2x^3}{(x-2)^2}$$

$$f'(x)=\frac{(x-2)^2\,6x^2-2x^3\times 2(x-2)}{(x-2)^4}$$

$$=\frac{2x^2(x-2)\left[3(x-2)-2x\right]}{(x-2)^4}$$

$$=\frac{2x^2(x-2)(x-6)}{(x-2)^4}$$

$$=\frac{2x^2(x-6)}{(x-2)^3}\quad x\neq 2$$

$$\therefore f'(x)=\frac{2x^2(x-6)}{(x-2)^3}=0 \text{ for SPs}$$

$$f'(x)=0\ \Rightarrow x=0,\quad x=6$$

SPs $(0,0),(6,27)$. **2A**

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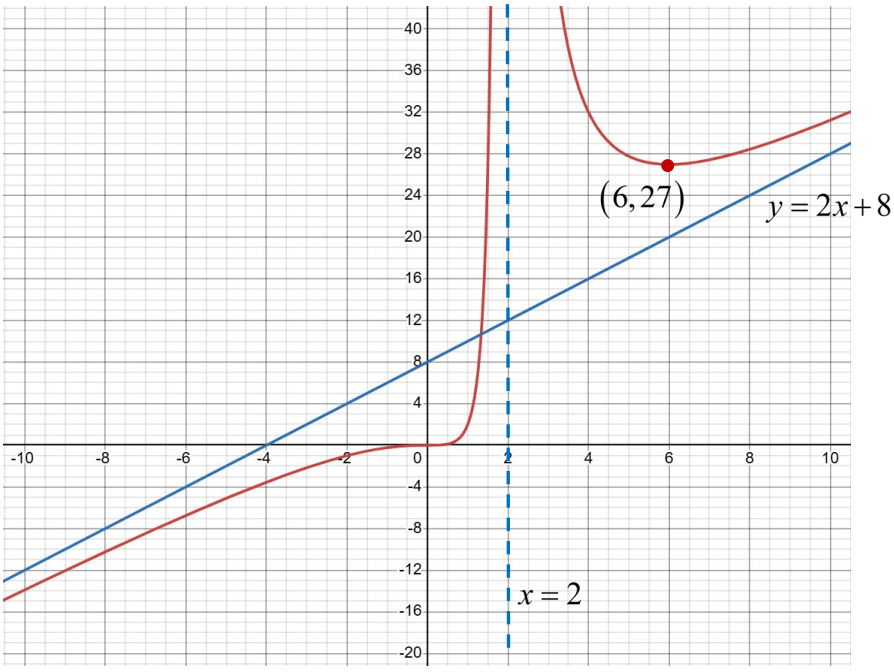
d. Sketch graph of $y = f(x)$.



1A asymptotes with equations
1A left branch correct shape
1A right branch correct shape including SP

e. Find the area under the graph of $f(x)$ between $x = 0$ and $x = 1$

Area $= \int_0^1 \left(2x + 8 + \frac{24x - 32}{(x-2)^2} \right) dx$

Partial Fractions

$\frac{24x - 32}{(x-2)^2} = \frac{A}{(x-2)^2} + \frac{B}{(x-2)}$

$24x - 32 = A + B(x-2)$

Let $x = 2 \Rightarrow 48 - 32 = A$

$\therefore A = 16$

Let $x = 0 \Rightarrow -32 = A - 2B$

$-32 = 16 - 2B$

$\therefore 2B = 48$

$\therefore B = 24$

$\frac{24x - 32}{(x-2)^2} = \frac{16}{(x-2)^2} + \frac{24}{(x-2)}$ **1M**

Area $= \int_0^1 \left(2x + 8 + \frac{24x - 32}{(x-2)^2} \right) dx = \int_0^1 \left(2x + 8 + \frac{16}{(x-2)^2} + \frac{24}{(x-2)} \right) dx$ **1M**

$= \int_0^1 \left(2x + 8 + 16(x-2)^{-2} + \frac{24}{(x-2)} \right) dx = \left[x^2 + 8x - \frac{16}{x-2} + 24\log_e(|x-2|) \right]_0^1$

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$= \left(1+8+16+24\log_e\left(|-1|\right)\right)-\left(0+8+24\log_e\left(|-2|\right)\right)$

$= 17+24\log_e\left(\frac{1}{2}\right)$

OR

Area $== 17-24\log_e(2)$ **1A**

Question 5 (3 marks)

Contradiction statement: $\sin(x)+\cos(x)>\sqrt{2}$ **1M**

$\left(\sin(x)+\cos(x)\right)^2>2$

$\sin^2(x)+\cos^2(x)+2\sin(x)\cos(x)>2$

$1+\sin(2x)>2$

$\sin(2x)>1$ **1M**

$\sin(2x)>1$ is impossible as $-1\le\sin(2x)\le 1$

$\therefore \sin(x)+\cos(x)\le\sqrt{2}$ **1A**

<u>Alternative solution</u>:

Contradiction statement: $\sin(x)+\cos(x)>\sqrt{2}$ **1M**

$\frac{1}{\sqrt{2}}\sin(x)+\frac{1}{\sqrt{2}}\cos(x)>1$

$\sin(x)\cos\left(\frac{\pi}{4}\right)+\cos(x)\sin\left(\frac{\pi}{4}\right)>1$

$\sin\left(x+\frac{\pi}{4}\right)>1$ **1M**

$\sin\left(x+\frac{\pi}{4}\right)>1$ is impossible as $-1\le\sin\left(x+\frac{\pi}{4}\right)\le 1$.

$\therefore \sin(x)+\cos(x)\le\sqrt{2}$ **1A**

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Question 6 (3 marks)

$P \sim N\left(a, b^2\right)$, with a and b defined in dollars (\$).

a. Let $X =$ daily sales from a store and $Y =$ daily running costs of a store.

$X \sim \mathrm{N}\left(1000, 60^2\right)$ and $Y \sim \mathrm{N}\left(400, 25^2\right)$

$P = \left(X_1 + X_2 + X_3 + X_4\right) - \left(Y_1 + Y_2 + Y_3 + Y_4\right)$ **1M**

$\mathrm{E}\left(P\right) = 4 \times 1000 - 4 \times 400 = 2400$

$$\begin{aligned}\mathrm{Var}\left(P\right) &= 4 \times 60^2 + 4 \times 25^2 \\ &= 2^2 \times 5^2 \times 12^2 + 2^2 \times 5^2 \times 5^2 \\ &= 2^2 \times 5^2 \times \left(12^2 + 5^2\right) \\ &= 2^2 \times 5^2 \times 13^2\end{aligned}$$

$\mathrm{sd}\left(P\right) = 130$

$\therefore a = 2400$ and $b = 130$ (a and b defined in dollars (\$)) **1A**

b.

$$\begin{aligned}\Pr\left(P > 2595\right) &= \Pr\left(Z > \frac{2595 - 2400}{130}\right) \\ &= \Pr\left(Z > \frac{195}{130}\right) \\ &= \Pr\left(Z > 1.5\right) \\ &= \frac{1 - 0.8664}{2} = \frac{0.1336}{2} = 0.0668\end{aligned}$$

1A

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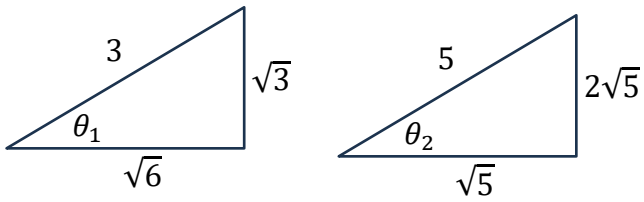
Question 7 (3 marks)

Given that $x = \sin^{-1}\left(\frac{\sqrt{3}}{3}\right) + \cos^{-1}\left(\frac{\sqrt{5}}{5}\right)$, where $x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

Find $\cos(2x)$.

Let, $\theta_1 = \sin^{-1}\left(\frac{\sqrt{3}}{3}\right)$ and $\theta_2 = \cos^{-1}\left(\frac{\sqrt{5}}{5}\right)$



$\sin(\theta_1) = \frac{\sqrt{3}}{3}$ and $\cos(\theta_2) = \frac{\sqrt{5}}{5}$

$\cos(\theta_1) = \frac{\sqrt{6}}{3}$ and $\sin(\theta_2) = \frac{2\sqrt{5}}{5}$ **1M**

Now, $x = \theta_1 + \theta_2$

$$\sin(x) = \sin(\theta_1 + \theta_2)$$
$$= \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2)$$
$$= \frac{\sqrt{3}}{3} \times \frac{\sqrt{5}}{5} + \frac{\sqrt{6}}{3} \times \frac{2\sqrt{5}}{5} = \frac{\sqrt{15}}{15} + \frac{2\sqrt{2} \times \sqrt{15}}{15} = \frac{\sqrt{15}}{15}\left(1 + 2\sqrt{2}\right)$$ **1M**

$$\sin^2(x) = \frac{1}{15}\left(1 + 2\sqrt{2}\right)^2 = \frac{1}{15}\left(1 + 4\sqrt{2} + 8\right) = \frac{1}{15}\left(9 + 4\sqrt{2}\right)$$

$$\cos(2x) = 1 - 2\sin^2(x) = 1 - 2 \times \frac{1}{15}\left(9 + 4\sqrt{2}\right) = \frac{15 - 18 - 8\sqrt{2}}{15} = \frac{-3 - 8\sqrt{2}}{15}$$ **1A**

Answer in form $\frac{a + b\sqrt{c}}{d}$, where $a, b, c, d \in Z$.

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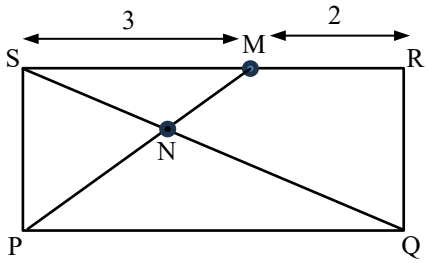
Question 8 (4 marks)

a.



$$\overrightarrow{PM} = \overrightarrow{PS} + \overrightarrow{SM}$$

$$= \overrightarrow{QR} + \frac{3}{5}\overrightarrow{SR}$$

$$= \overrightarrow{QR} + \frac{3}{5}\overrightarrow{PQ}$$

$$= 6\underset{\sim}{i} + 4\underset{\sim}{j} + \frac{3}{5}\left(-5\underset{\sim}{i} + 10\underset{\sim}{j}\right)$$

$$= 3\underset{\sim}{i} + 10\underset{\sim}{j}$$ **1A**

b. $\overrightarrow{PN} = m\overrightarrow{PQ} + (1-m)\overrightarrow{QR}$

$$= m\left(-5\underset{\sim}{i} + 10\underset{\sim}{j}\right) + (1-m)\left(6\underset{\sim}{i} + 4\underset{\sim}{j}\right)$$

$$= (6 - 11m)\underset{\sim}{i} + (4 + 6m)\underset{\sim}{j}$$ **1M**

Also, $\overrightarrow{PN} = n\overrightarrow{PM}, n \in R$

$$= n\left(3\underset{\sim}{i} + 10\underset{\sim}{j}\right)$$

So, $6 - 11m = 3n \Rightarrow 11m + 3n = 6$ (1)

$4 + 6m = 10n \Rightarrow 3m - 5n = -2$ (2) **1M**

From $(1) \times 5 + (2) \times 3$,

$$55m + 9m = 30 - 6$$

$$64m = 24$$

$$\therefore m = \frac{3}{8}$$ **1A**

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Question 9 (4 marks)

a.

Normal of plane $\Pi_1, \underset{\sim}{n}_1 = 2\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}$

$\underset{\sim}{u}.\underset{\sim}{n}_1 = 0 \Rightarrow 2a - 2b - 6 = 0 \Rightarrow a - b = 3$ (1)

$\underset{\sim}{v}.\underset{\sim}{n}_1 = 0 \Rightarrow 2 + (a+1) - (2-b) = 0 \Rightarrow a + b = -1$ (2) **1M**

From $(1)+(2), 2a = 2 \Rightarrow a = 1$

From $(1), 1 - b = 3 \Rightarrow b = -2$

$\therefore a = 1, b = -2$ **1A**

b. Vector equation of the line of intersection of the planes Π_1 and Π_2.

Normal of plane $\Pi_1, \underset{\sim}{n}_1 = 2\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}$

Normal of plane $\Pi_2, \underset{\sim}{n}_2 = \underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}$

Direction vector of intersection line, $\underset{\sim}{d} = \underset{\sim}{n}_1 \times n_2 = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 2 & 1 & -1 \\ 1 & -2 & 1 \end{vmatrix} = -\underset{\sim}{i} - 3\underset{\sim}{j} - 5\underset{\sim}{k}$ **1M**

A common point on both planes:

$\Gamma_1 : 2x + y - z = 10$ and $\Gamma_2 : x - 2y + z = 5$

Let, $y = 0$,

$2x - z = 10$ (1)

$x + z = 5$ (2)

From $(1)+(2), 3x = 15 \Rightarrow x = 5$

From $(2), 5 + z = 5 \Rightarrow z = 0$

Many possible points

A point on the intersection line is $(5,0,0)$

Line equation, $\underset{\sim}{r}(t) = \boxed{(5,0,0)} + t(-1,-3,-5), t \in R.$

Or, $\underset{\sim}{r}(t) = \boxed{5\underset{\sim}{i}} + t(-\underset{\sim}{i} - 3\underset{\sim}{j} - 5\underset{\sim}{k}), t \in R.$ **1A**

Line equation will change with chosen point.

Many possible correct answers

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Question 10 (3 marks)

Evaluate $\int_0^{\frac{\pi}{2}} \left(2x^2 \cos^2(2x+\pi)\right) dx$.

Solution:

Let, $I = \int 2x^2 \cos^2(2x+\pi)\, dx$

$2\cos^2(2x+\pi) = 1 + \cos(4x+2\pi) = 1 + \cos(4x)$

So, $I = \int x^2 + x^2 \cos(4x) dx = \frac{x^3}{3} + \int x^2 \cos(4x) dx$

Let, $I_1 = \int x^2 \cos(4x)\, dx$

Integration by parts,

$u = x^2 \;\Rightarrow u' = 2x$

$v' = \cos(4x) \;\Rightarrow v = \int \cos(4x)\, dx = \frac{1}{4}\sin(4x)$

$I_1 = x^2 \times \frac{1}{4}\sin(4x) - \int 2x \times \frac{1}{4}\sin(4x)\, dx = \frac{x^2}{4}\sin(4x) - \frac{1}{2}\int x \times \sin(4x)\, dx$ **1M**

Let, $I_2 = \int x \times \sin(4x)\, dx$

Integration by parts,

$f = x \;\Rightarrow f' = 1$

$g' = \sin(4x) \;\Rightarrow g = \int \sin(4x)\, dx = -\frac{1}{4}\cos(4x)$

$I_2 = x \times -\frac{1}{4}\cos(4x) - \int 1 \times -\frac{1}{4}\cos(4x)\, dx = -\frac{x}{4}\cos(4x) + \frac{1}{4}\int \cos(4x)\, dx = -\frac{x}{4}\cos(4x) + \frac{1}{16}\sin(4x)$

So, $I_1 = \frac{x^2}{4}\sin(4x) - \frac{1}{2}\left(-\frac{x}{4}\cos(4x) + \frac{1}{16}\sin(4x)\right) = \frac{x^2}{4}\sin(4x) + \frac{x}{8}\cos(4x) - \frac{1}{32}\sin(4x)$

So, $I = \frac{x^3}{3} + \frac{x^2}{4}\sin(4x) + \frac{x}{8}\cos(4x) - \frac{1}{32}\sin(4x) + c$ **1M**

Now, $\int_0^{\frac{\pi}{2}} \left(2x^2 \cos^2(2x+\pi)\right) dx = \left[\frac{x^3}{3} + \frac{x^2}{4}\sin(4x) + \frac{x}{8}\cos(4x) - \frac{1}{32}\sin(4x)\right]_0^{\frac{\pi}{2}}$

$= \left[\frac{\pi^3}{3\times 8} + 0 + \frac{\pi}{8\times 2} - 0\right] - 0$

$= \frac{\pi^3}{24} + \frac{\pi}{16}$ **1A**

End of Solutions

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